Literature Map: Wave-Energy Forecasting, Uncertainty, and Storage-Aware Smoothing

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Source notebook: 00_literature_map.ipynb

00 — Literature Map: Wave-Energy Forecasting, Uncertainty, and Storage-Aware Smoothing

1. Notebook purpose

This notebook maps the literature motivation behind this project. It is not intended to be a full literature review. Instead, it explains how work on battery energy storage, renewable-power smoothing, wave-energy forecasting, prediction intervals, and hybrid storage systems motivates the workflow used in this repository.

The notebook helps answer:

1. Why wave-energy forecasting matters.
2. Why uncertainty and prediction intervals matter.
3. Why storage-aware smoothing is relevant for wave energy.
4. Why a simplified WEC power-estimation workflow is acceptable for this project.
5. How this repository connects forecasting and uncertainty to storage-aware smoothing metrics relevant to BESS/HESS grid-integration questions.

2. Personal and technical motivation

My interest in wave energy started during my Bachelor of Engineering (BEng) in Environmental Engineering, where I became interested in marine renewable energy as part of the wider clean-energy transition. Later, during my Master of Science (MSc) in Computing Sciences – Data Science, this interest became more focused on how data-driven modelling, forecasting, uncertainty estimation, and reproducible workflows can support renewable-energy analysis.

This project connects those two directions. It starts from open wave-resource data, estimates a simplified WEC power signal, benchmarks short-term forecasting methods, adds uncertainty estimates, and finally links the results to storage-aware smoothing metrics relevant to BESS and HESS applications.

My prior portfolio work on battery diagnostics from impedance data also reflects the same motivation: using technical data preparation, feature engineering, model benchmarking, and engineering interpretation in energy-storage applications.

3. Why wave energy and storage-aware forecasting?

Wave energy is still an emerging renewable-energy technology, but official targets and funding programmes show continuing interest in marine renewable energy. The European Union has set an objective of at least 1 GW of installed ocean-energy capacity by 2030 and 40 GW by 2050 (European Commission, n.d.). In the United States, the Department of Energy describes marine energy from waves, tides, and river or ocean currents as a large renewable resource, while recent funding calls continue to support marine-energy research and development (U.S. Department of Energy, n.d.; U.S. Department of Energy, 2024).

From a data-science perspective, wave energy is interesting because WEC power output depends on changing sea states and can fluctuate strongly over time. For grid integration, this makes point forecasts alone insufficient: the uncertainty around those forecasts and the resulting power ramps are also important.

This motivates a workflow that connects wave-resource data, simplified WEC power estimation, short-term forecasting, prediction intervals, and storage-aware smoothing metrics.

4. Literature map table

The literature map below organizes the papers and technical sources used to motivate this project. The selected sources touch on battery energy storage, renewable-power smoothing, wave-energy forecasting, uncertainty estimation, WEC output smoothing, and hybrid storage systems.

The table is intended as a compact reference map rather than a full literature review. It highlights the main topic of each source, the type of data or method used, and its relevance to the broader project direction.

No.	Ref.	Title	Theme	Method	Data	Relevance
1	Lundmark et al. (2026)	A validated forecasting model for optimizing dispatch schedules of PV-battery system in grid-connected apartment buildings	BESS dispatch and PV integration	Forecast-informed dispatch scheduling	Measured PV and apartment electricity data with grid-connected BESS operation	Motivates the link between forecasts, storage operation, self-consumption, and grid interaction.
2	Bobček et al. (2025)	A Novel Approach to Day-Ahead Forecasting of Battery Discharge Profiles in Grid Applications Using Historical Daily Profiles	BESS forecasting and grid applications	Forecasting battery discharge profiles	Historical measured BESS discharge profiles in grid applications	Supports the broader data-science framing of benchmarking forecasting methods for energy-storage-related time series.
3	Shabani and Yan (2025)	Optimization and evaluation of operational and economic performance of grid-connected battery storage at different stages of battery health	BESS economics and battery health	Operation and economic optimization under battery-health constraints	Grid-connected BESS operation and optimization study	Motivates why storage operation should not be treated as independent of battery health and operating objectives.

No.	Ref.	Title	Theme	Method	Data	Relevance
4	Liu et al. (2025)	A Framework for Anomaly Cell Detection in Energy Storage Systems Based on Daily Operating Voltage and Capacity Increment Curves	BESS safety and health monitoring	Cell anomaly detection	Measured or operational ESS cell-monitoring data	Lightly motivates the safety and maintenance importance of BESS monitoring, without making diagnostics part of this project.
5	Ji et al. (2025)	Variational Autoencoder Based Anomaly Detection in Large-Scale Energy Storage Power Stations	BESS safety and health monitoring	VAE-based anomaly detection	Large-scale energy-storage power-station monitoring data	Reinforces that BESS operation involves reliability and monitoring concerns beyond simple energy buffering.
6	Qi et al. (2025)	Remaining Available Energy Prediction for Energy Storage Batteries Based on Interpretable Generalized Additive Neural Network	BESS prediction and interpretability	Remaining available energy prediction	Battery operation and feature-extraction data	Supports the idea that interpretable prediction models can help energy-storage operation and reliability.
7	Figgener et al. (2024)	Multi-year field measurements of home storage systems and their use in capacity estimation	Battery field measurements and capacity estimation	Capacity estimation from long-term field measurements	Multi-year measured home-storage field data	Motivates why real storage operation and capacity fade matter when considering renewable integration.
8	Angenendt et al. (2016)	Enhancing Battery Lifetime in PV Battery Home Storage System Using Forecast Based Operating Strategies	Forecast-based BESS operation	Forecast-based operating strategies for battery lifetime	PV home-storage operation study with forecast-based control	Motivates the general idea that forecasts can improve storage operation, lifetime, and renewable self-consumption.

No.	Ref.	Title	Theme	Method	Data	Relevance
9	Gräf et al. (2022)	What drives capacity degradation in utility-scale battery energy storage systems? The impact of operating strategy and temperature in different grid applications	Utility-scale BESS degradation	Capacity degradation under different grid applications	Utility-scale BESS operating-strategy and temperature modelling study	Motivates keeping battery degradation in mind while avoiding detailed degradation modelling in this project.
10	Naumann (2018)	Techno-economic evaluation of stationary lithium-ion energy storage systems with special consideration of aging	Stationary storage techno-economics	Techno-economic BESS simulation with aging	Stationary lithium-ion storage modelling and application scenarios	Provides broader BESS context, including grid-service and aging-aware stationary storage applications.
11	Möller et al. (2022)	SimSES: A holistic simulation framework for modeling and analyzing stationary energy storage systems	Stationary storage simulation	Open-source BESS simulation framework	Software/framework for simulated stationary energy-storage systems	Provides background for possible future BESS simulation and reinforces why storage operation can be modelled explicitly
12	Fontana Crespo et al. (2024)	A comparative analysis of Machine Learning Techniques for short-term grid power forecasting and uncertainty analysis of Wave Energy Converters	WEC forecasting and uncertainty	Short-term grid-power forecasting and prediction intervals	Simulated WEC grid-power data	Directly motivates forecasting benchmarks, forecast horizons, and prediction-interval evaluation for WEC power.
13	Mérigaud and Ringwood (2018)	Power production assessment for wave energy converters: Overcoming the perils of the power matrix	WEC power estimation	Power-matrix approach and its limitations	Research paper on WEC power production assessment	Supports using a power-matrix-style approximation while clearly motivating its limitations

No.	Ref.	Title	Theme	Method	Data	Relevance
14	Klise et al. (2020)	MHKiT (Marine and Hydrokinetic Toolkit) - Python	Marine-energy data processing	Open-source Python toolkit for marine renewable energy data and performance analysis	Official software reference for wave-resource and device-performance workflows	Supports a Python-first implementation using established marine-energy data-processing tools
15	Blanco et al. (2025)	Power Smoothing in a Wave Energy Conversion Using Energy Storage Systems: Benefits of Forecasting-Enhanced Filtering for Reduction in Energy Storage Requirements	WEC power smoothing with ESS	Forecast-enhanced filtering and storage requirement reduction	WEC simulation driven by wave or weather conditions with ESS smoothing analysis	Directly motivates connecting WEC power forecasts to storage-aware smoothing metrics.
16	Pelosi et al. (2024)	A Hybrid Energy Storage System Integrated with a Wave Energy Converter: Data-Driven Stochastic Power Management for Output Power Smoothing	WEC smoothing with HESS	Battery-supercapacitor stochastic power management	Simulated WEC output and hybrid storage smoothing study	Motivates the BESS/HESS framing and the idea that fast WEC power fluctuations may be better handled with hybrid storage.

5. Main literature themes

The sources in the literature map fall into three main themes. First, battery energy storage studies motivate the broader grid-integration context. This includes dispatch optimization, profitability, grid services, battery degradation, field capacity estimation, and safety-oriented monitoring. A central motivation from this literature is that BESS operation is not only an energy-balancing problem: the chosen operating strategy can also affect battery aging, maintenance needs, and long-term techno-economic performance.

Second, WEC forecasting studies motivate the forecasting and uncertainty parts of the project. Short-term WEC grid-power forecasting and uncertainty analysis are especially relevant because forecast horizons, model comparison, and prediction intervals can support different operational decisions (Fontana Crespo et al., 2024).

Third, WEC smoothing and hybrid-storage studies motivate the storage-aware part of the workflow. WEC output smoothing using storage can support grid-connected operation by reducing the impact of power fluctuations. In this project, that literature motivates storage-aware metrics such as ramp reduction, storage throughput, and indicative energy-buffer requirements, rather than a detailed BESS/HESS design study. Forecast-enhanced filtering can reduce storage requirements for WEC output smoothing (Blanco et al., 2025). Hybrid storage with batteries and supercapacitors is especially relevant because the supercapacitor can absorb faster power fluctuations, potentially reducing the stress that highly variable WEC output would otherwise impose on the battery (Pelosi et al., 2024).

Across these themes, the common thread is the use of technical data processing, prediction, model comparison, uncertainty, and engineering interpretation to study wave-energy, storage, and grid-integration problems.

6. How the literature maps to the workflow

The literature map motivates a workflow that moves from wave-resource data toward grid-relevant interpretation. The project first prepares open wave-resource data, then converts the sea-state time series into an estimated WEC power signal. This power signal is used for short-term forecasting, prediction-interval estimation, and storage-aware smoothing metrics.

Workflow part	Main purpose	Literature motivation
Wave-resource data preparation	Build a clean time series of sea-state variables	Wave-energy forecasting and WEC simulation studies rely on sea-state or WEC-output time series
Simplified WEC power estimation	Convert sea-state data into an approximate power signal	Power-matrix approaches provide a practical way to relate sea states to WEC output
Forecasting baselines	Predict short-term WEC power output	Forecast horizons and model comparison are central in WEC forecasting studies
Prediction intervals	Quantify forecast uncertainty	Prediction intervals help move beyond point forecasts
Storage-aware smoothing metrics	Connect power variability to storage relevance	ESS/HESS studies motivate ramp reduction, smoothing, and hybrid storage concepts

7. What this project does — and does not do

This project is intended as a lightweight, reproducible, Python-first workflow. It connects open wave-resource data with simplified WEC power estimation, forecasting baselines, uncertainty estimation, and storage-aware smoothing metrics. The aim is to demonstrate a clear data-science workflow for wave-energy and storage-related grid-integration questions.

Area	Included here?	Note
Open wave-resource data preparation	Yes	Clean and prepare sea-state time series for later analysis
Simplified WEC power estimation	Yes	Use a power-matrix-style approach as a practical approximation

Area	Included here?	Note
Short-term forecasting baselines	Yes	Compare simple, interpretable forecasting models
Prediction intervals / uncertainty	Yes	Add uncertainty information around point forecasts
Storage-aware smoothing metrics	Yes	Relate WEC power variability to storage-relevant smoothing metrics
Detailed WEC hydrodynamic simulation	No	Outside the scope of this lightweight workflow
Device-validated WEC power prediction	No	The estimated power signal is illustrative, not device-certified
Detailed battery degradation modelling	No	Battery aging motivates the project, but is not modelled directly
Cell-level battery diagnostics	No	Relevant for safety and maintenance, but outside the current workflow

The simplified WEC power-estimation step should therefore be interpreted carefully. Power-matrix-style methods are useful for reproducible performance estimation, but they can also hide important device-specific and sea-state-dependent effects if used uncritically (Mérigaud & Ringwood, 2018). In this project, the approach is treated as a practical approximation consistent with marine-energy toolchains such as MHKiT-Python and IEC-oriented WEC power-performance concepts (IEC, 2024). It does not replace detailed WEC hydrodynamic modelling, control-system simulation, or validation against a specific device.

8. Guiding questions

This project is guided by three practical questions:

1. How well do simple short-term forecasting baselines predict estimated WEC power output?
2. Can prediction intervals provide useful information about forecast uncertainty beyond point-forecast accuracy?
3. How can WEC power variability, forecast errors, and uncertainty be connected to storage-aware smoothing metrics relevant to BESS/HESS applications?

The earlier workflow steps, including wave-resource data preparation and simplified WEC power estimation, are treated as necessary foundations for answering these questions rather than as research questions by themselves.

9. References

European Commission. (n.d.). *Marine renewable energy*. European Commission: Oceans and Fisheries. https://oceans-and-fisheries.ec.europa.eu/ocean/blue-economy/marine-renewable-energy_en. Accessed: 23 June 2026.

U.S. Department of Energy. (n.d.). *Marine Energy Program*. Office of Critical Minerals and Energy Innovation. https://www.energy.gov/cmei/water/marine-energy-program. Accessed: 23 June 2026.

U.S. Department of Energy. (2024). *U.S. Department of Energy invests more than $18 million in marine energy and offshore wind research at universities.* Office of Critical Minerals and Energy Innovation. https://www.energy.gov/cmei/articles/us-department-energy-invests-more-18-million-marine-energy-and-offshore-wind-research. Accessed: 23 June 2026.

Fontana Crespo, A., et al. (2024). *A comparative analysis of Machine Learning Techniques for short-term grid power forecasting and uncertainty analysis of Wave Energy Converters.* Engineering Applications of Artificial Intelligence. https://doi.org/10.1016/j.engappai.2024.109352

Blanco, M., et al. (2025). *Power smoothing in a wave energy conversion using energy storage systems: Benefits of forecasting-enhanced filtering.* Applied Sciences. https://doi.org/10.3390/app152011106

Pelosi, D., et al. (2024). *A Hybrid Energy Storage System Integrated with a Wave Energy Converter: Data-Driven Stochastic Power Management for Output Power Smoothing.* Energies, 17(5), 1167. https://doi.org/10.3390/en17051167

International Electrotechnical Commission. (2024). *IEC TS 62600-100:2024: Marine energy — Wave, tidal and other water current converters — Part 100: Electricity producing wave energy converters — Power performance assessment.* IEC. https://webstore.iec.ch/en/publication/66061

Klise, K., Pauly, R., Ruehl, K. M., Olson, S., Shippert, T., Morrell, Z., Bredin, S., Lansing, C., Macduff, M., Martin, T., Sivaraman, C., Gunawan, B., & Driscoll, F. (2020). *MHKiT (Marine and Hydrokinetic Toolkit) - Python* [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.3924683

Mérigaud, A., & Ringwood, J. V. (2018). Power production assessment for wave energy converters: Overcoming the perils of the power matrix. Proceedings of the Institution of Mechanical Engineers, Part M: Journal of Engineering for the Maritime Environment, 232(1), 50–70. https://doi.org/10.1177/1475090217731671